

Project Initialization and Planning Phase

Date	06 July 2026
Team ID	
Project Title	Voice Based Concept Understanding Analyser
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	To evaluate users' conceptual understanding and speech clarity using AI-based voice analysis and provide automated feedback.
Scope	The system analyzes spoken explanations using speech-to-text, NLP, and audio processing to assess understanding and communication skills.
Problem Statement	
Description	Traditional evaluation methods cannot effectively measure conceptual understanding and speaking skills through voice explanations.
Impact	This leads to inaccurate assessment, lack of feedback, and difficulty in improving communication and learning outcomes.
Proposed Solution	
Approach	The system uses AI models to convert speech to text, analyze semantic meaning, and evaluate speech fluency to generate scores and feedback.
Key Features	Speech-to-text conversion using Whisper Semantic analysis using Sentence-BERT

	Fluency analysis using Librosa Automated scoring and feedback generation Downloadable PDF reports
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	Standard CPU (no GPU required)	
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	10–20 GB for audio files and reports
Software		
Frameworks	Streamlit	
Libraries	Whisper, Transformers, Librosa, NLTK, Matplotlib	
Development Environment	VS Code / Jupyter Notebook	
Data		
Data	Audio input from users and predefined reference text for concept comparison	